

No Tutorial Today

Tutorials start on Sep 24.

1.1 Linear Equations and Inequalities in One Variable

An equation of the form

$$ax + b = 0 \tag{1}$$

is called a *linear equation in one variable*, where a, b are real numbers and $a \neq 0$, and x is a *variable*.

To solve equation (1), we move the constant term b to the right hand side (RHS) and rewrite it as

$$ax = -b. \tag{2}$$

We then divide both sides of (2) by a to obtain

$$x = \frac{-b}{a}.$$

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Example The equation

$$3x + 6 = 0 \tag{3}$$

is solved by the following steps:

$$\begin{aligned} 3x &= -6 \\ x &= \frac{-6}{3} \\ &= -2. \end{aligned}$$

Hence $x = -2$ is the solution to equation (3).

In the above steps, we used the following properties about equations.

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Property 1: Adding (or subtracting) the same number on both sides of an equation.

$$A = B \iff A + C = B + C \iff A - C = B - C.$$

Hence

$$A = B \iff A - B = 0.$$

This means one can move a term from one side to the other by changing its sign.

Property 2: Multiplying (or dividing) both sides of an equation by a *nonzero number*. For $C \neq 0$ we have

$$A = B \iff AC = BC \iff \frac{A}{C} = \frac{B}{C}.$$

Remark Different names can be used for the variable in an equation. For example, $5v + 10 = 0$ and $\frac{1}{2}t - 3 = 0$ are all linear equations.

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The variable may appear on both sides of an equation. For example,

$$7(x + 1) = 2(2x - 1) \quad (4)$$

is also a linear equation, because we can rewrite it in the form of (1) or (2) using the following steps:

Step 1 Use the distributive law to expand $7(x + 1) = 7x + 7$ and $2(2x - 1) = 4x - 2 \implies 7x + 7 = 4x - 2$.

Step 2 Move the x -term to LHS, and all constant terms to RHS.
 $7x - 4x = -2 - 7 \implies 3x = -9$.

Remember to change the sign of the term when it is moved from one side to the other side of the equation.

Step 3 Dividing both sides by the coefficient of x to obtain the solution $x = \frac{-9}{3} = -3$.

The Absolute Value Equation

Check the answer: $\text{LHS} = 7(-3 + 1) = 7 \times (-2) = -14$,
 $\text{RHS} = 2(2 \times (-3) - 1) = 2 \times (-7) = -14$.

Recall that a real number corresponds to a point in the real line. The *absolute value* of x , denoted by $|x|$, is the distance from the point x in the real line to the origin. Hence

$|x| = x$ when $x \geq 0$, and $|x| = -x$ when $x < 0$.

Examples: $|-3| = -(-3) = 3$. $|0.1| = 0.1$, $|0| = 0$.

Suppose $a > 0$, then the equation $|x| = a$ has two solutions:
 $x = a$ or $x = -a$.

The equation $|x| = 0$ has one solution $x = 0$.

We note that the absolute value equation is *not* a linear equation.

Example Solve the equation $|2x - 5| = 1$.

Solution We have $2x - 5 = 1$ or $2x - 5 = -1$. Hence
 $x = \frac{6}{2} = 3$ or $x = \frac{4}{2} = 2$.

Check: $|2 \times 3 - 5| = |1| = 1$ and $|2 \times 2 - 5| = |-1| = 1$.

1.1 Linear Equations and Inequalities in One Variable

A *linear inequality in one variable* is an inequality that can be written in one of the following forms:

$$ax + b < 0, \quad ax + b \leq 0, \quad ax + b > 0, \quad ax + b \geq 0,$$

where a and b are real numbers and $a \neq 0$.

Property 1: Adding (or subtracting) the same number on both sides of an inequality leaves the inequality unchanged.

$$a < b \iff a + c < b + c, \quad a - c < b - c,$$

Property 2: Multiplying (or dividing) both sides of an inequality by a *positive* number leaves the inequality unchanged.

For $c > 0$ we have $a < b \iff ac < bc, \quad \frac{a}{c} < \frac{b}{c}.$

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Property 3: Multiplying (or dividing) both sides of an inequality by a *negative* number changes the direction of the inequality.

For $c < 0$ we have $a < b \iff ac > bc, \quad \frac{a}{c} > \frac{b}{c}.$

The above three properties also hold for other three types of inequalities.

Example Solve the linear inequality $5x > 10$.

Solution Dividing both sides by 5, we obtain $x > \frac{10}{5}$. That is, $x > 2$.

Example Solve the linear inequality $-5x > 10$.

Solution Dividing both sides of the original inequality by -5 : $x < \frac{10}{-5}$, that is, $x < -2$.